Interfacing for dynamic morphology in computer music performance

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Abstract

Laptop performance of computer music has become wide spread in the electronic music community. It brings with it many issues pertaining to the communication of musical intent. Critics argue that performances of this nature fail to engage audiences as many performers use the mouse and keyboard to control their musical works, leaving no visual cues to guide the audience as to the correlation between performance gestures and musical outcomes. Interfaces need to communicate something of their task. The author will argue that cognitive affordances associated with the performance interface become paramount if the musical outcomes are to be perceived as clearly tied to realtime performance gestures, ie. That the audience is witnessing the creation of the music in that moment as distinct to the manipulation of pre-recorded or pre-sequenced events.

Introduction

Acoustic Instrument Model

In traditional musical performance, the interface has always been part of the instrument - part of the excitation-sonification system. For instance, all instruments in the string family share a mechanism for retaining the string which sees the string attached to the tail piece, run over the bridge, which sits on the sounding board, (dispersing and amplifying the vibrations of the string) and runs to the peg, held in the scroll allowing for variations in tuning. This mechanism provides for multiple independent sound sources (four or five strings) to be held on a single instrument body. Whilst this point may seem trivial, it allows a single musician to execute multiple parallel musical ideas on a single instrument, a techniques well illustrated in the solo cello suits by J.S Bach or the solo violin works of Paganini, although in reality, the design of the instrument only allows two notes to be sounded simultaneously, with three or four note patterns sounding in quick succession (appearing almost simultaneous).

Recent work carried out by the author (Paine, Stevenson, & Pearce, 2007) at the University of Western Sydney examined the fundamental control parameters utilised by expert musicians on traditional instruments. The resultant model (see Figure 1) presents the musical parameters; Dynamics, Pitch, Vibrato, Articulation and Attack/release as the focus of the physical instrument control, and of primary focus in achieving a well developed instrumental tone, the principle concern for all interviewed musicians. The notion of musical expression, or musicianship plays a role in aesthetic decisions, and acts as an overall metric for the musical sensibility that underlies all instrumental training and musical decision-making. The above musical parameters were then considered in terms of the physical control musicians employ in order to achieve musical outcomes (see Figure 2). For instance one subject commented that " pitch is controlled by the position of the bow, as well as the movement of the bow between the bridge and the fingerboard", a flute player commented that a "fast air stream produces a dark tone, whilst a relaxed, slow air stream produces a lighter, softer tone. The higher pitch registers are achieved by using a faster air stream, angled slightly upwards,

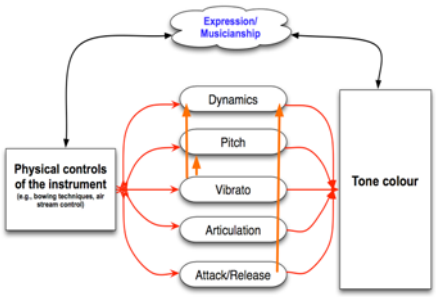


Figure 1. ***Musical parameter overview***

whilst the lower registers require a slower downward air stream". An analysis of such statements lead to the identification of four primary physical controls; pressure, speed, angle and position (Figure 2).

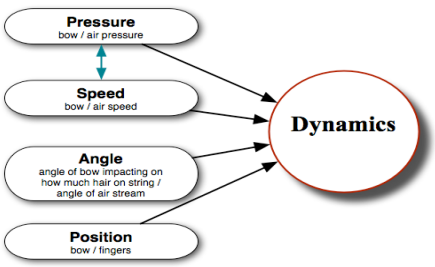


Figure 2 *Control parameter model*

Applying the Model in Computer Music

In computer music, the excitation-sonification relationship is broken into interface and synthesis algorithm.

The selection of an interface that provides the above defined physical control parameters; pressure, speed, angle and position, and the careful mapping of the parameter space can encourage a link between generation (gestural quality – intense, fast, light etc) and musical outcome. The author has been experimenting with the application of the above physical control parameter model for electronic music performance using experimental interfaces such as the Wacom Graphics Tablet and the Nintendo Wii Remote (WiiMote). This discussion will focus on the WiiMote.

Nintendo Wii Remote

The Nintendo Wii Remote (WiiMote) has been the focus of an explosion of music performance, DJ and VJ experimentation. Software frameworks quickly appeared for the Macintosh computer and the Windows and Linux operating systems, and were compiled into objects or plugins for programming languages such as Max/MSP, Quartz Composer (QCWii), Isadora, OSCulator (providing an OSC bridge) and were embedded in applications such as WiiToMIDI, Wiinstrument and Wii Loop Machine (actually developed in Max/MSP)

WiiMote: X-axis accel, X-axis accel, X-axis accel, Pitch, Roll, Yaw, 2 or 3 Buttons, WiiBarX, WiiBarY

Nunchuck: X-axis accel, X-axis accel, X-axis accel, Pitch, Roll, Yaw, X/Y of joystick, 2 Buttons

Table 1 WiiMote and Nunchuck data set

Part of the attraction will have been that the interface is wireless (Bluetooth), but a major factor will also have been the range of control afforded by the WiiMote. The WiiMote contains eleven buttons (momentary) and a three-dimensional accelerometer (6 continuous streams of data – see Table 1 below). In addition to this already substantial data set, the WiiMote accepts an accessory called a Nunchuck, which is held in the remaining free hand and contains a second three-dimensional accelerometer, a traditional two-dimensional joystick and two trigger style buttons (C and Z).

One of the characteristics of such an interface is that the buttons and or the joystick can be used in parallel with, and independent of the three-dimensional accelerometer. This means that the following data streams can be performed semi-independently and simultaneously.

An infrared Sensor Bar (WiiBar) can be added to this setup to sense the absolute position of the WiiMote on the X and Y-axis, providing an additional two continuous data element. Table 1 indicates that the WiiMote can produce up to nine simultaneous but partly inter-related data streams (6 continuous and 3 momentary) and the Nunchuck can produce up to ten simultaneous but partly inter-related data streams (8 continuous and 2 momentary). When combined with the WiiBar a data set of 16 continuous and 5 momentary elements can be produced.

This represents too many pieces of data for an individual to use constructively at any one time. In order to consider the WiiMote/ Nunchuck combination against the acoustic instrument model outlined above, an analysis of the parameter space was examined against the characteristics of pressure, speed, angle and position.

Pressure: XAccel, YAccel, ZAccel, XJoy, YJoy. Pitch, Roll and Yaw may all be interpreted in this way as they provide a continuous data stream.

Speed: can be calculated from all continuous data streams

Angle: Pitch, Roll and Yaw

Position: Pitch, Roll and Yaw, NunJoyX/Y, WiiBarX/Y and all buttons (11 with up to 5 simultaneously)

As the continuous data streams are simultaneous but partly inter-related, some decisions need to be made regarding the use of control parameters in synthesis where by their musical outcomes could conflict. For instance the X/Y/Z Accel data streams cannot be independent – a movement in any one axis will cause some variation in the other axis. Pitch, Roll and Yaw may also be affected by accelerative movements in the X/Y/Z planes, however within themselves they present more independence that the acceleration data, which in all planes is subject to gravity. The buttons are of course independent although only momentary. The fundamental consideration in constraining/choosing the data set is an analysis of the independence or otherwise of the generation of the control data, guided by an understanding of the gestural qualities best suited to the timbral/pitch/temporal qualities of the composition, and the way in which these musical potentials could be navigated in performance.

It should also be noted at that this point that the more advanced Wacom tablets such as the Intuos3 provides at least thirteen pieces of continuous data and two buttons from a single pen (multiple pens may be independently mapped, but not used simultaneously) as is illustrated in the Wacom pen assignment page in the Kyma software (see Figure 3).

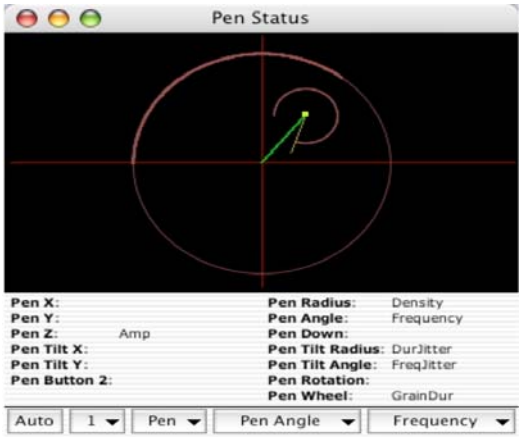


Figure 3 *Kyma software Wacom interface status screen*

Instrument Design

Once the controller parameters are understood, the instrument design begins. In addition to the above discussion, electronic music performance differs from acoustic in that the composer designs and builds the instruments to suit the demands of the composition. The gestural quality of interface parameter sets plays an important part in the associated gesture of the musical outcomes – slow or fast transitions, fast chaotic timbral changes, accuracy in pitch or amplitude and repeatability are just some of the considerations at hand. It should be noted that the composition, instrument design, interface design and mapping strategies apply constraints that define the musical work and the performance options. Instrument design requires some serious consideration of musical aesthetic and the relationship between performer and audience. The interface and its implementation therefore serve two primary goals: 1) to increase performability, allowing the musician to nuance musical outcomes in a way not possible with existing interfaces or using the mouse/keyboard computer interface; 2) To increase communication with the audience, displaying something of the energy and intent of the performer, providing a conduit for engagement in the realtime qualities of the performance – ie. The ritual of performance.

Any implementation of a new musical interface must therefore consider the ecology of this environment. Gurevich and Treviño (2007) discussed the development of a framework for an ecology of musical action:

> An ecological framework without the assumption of a commodity or a singular creator makes it admittedly difficult to unify or relate the experiences of the individual actors in the system. Donald Norman's (Norman, 2004) formulation of three levels of processing in the human brain and associated modes of experience facilitates a meaningfully descriptive but inclusive consideration of the musical experience from a variety of points of view. The three levels of processing are visceral, automatic and pre-wired reactions to sensory stimuli; behavioral, involved in the subconscious control of learned everyday actions (driving a car, taping, playing a violin); and reflective, the highest-level conscious thought in which we form opinions, plans, and abstractions. … While Norman argues that good design requires a balanced appeal on all three levels, it is also clear that all three levels are engaged in creating music. … Norman describes the skilled performer's ability to play a piece unconsciously (Behavioral) while simultaneously considering matters of the large-scale form (reflective). The listener reacts viscerally to the sound and may also contemplate meaning. (pg. 109)

Gurevich and Treviño point to the difficulty of identifying a unified source for musical creation in a complex system where the interface and the sonification mechanism are separated. Their adoption of the three levels of processing outlined by Donald Norman (Norman, 2004) illustrate some of the cognitive associations brought into play when engaging both as a performer and an audience member during a musical performance. The author supports Gurevich and Treviño statement that musicians are working at all three levels identified by Norman when improvising and performing dynamic scores. The visceral and behavioral levels are enshrined in the kinetic gesturing that brings about the musical outcomes, representing a sonification of the performative gesture. The reflective layer is brought to bear by the musician who is actively, but probably subconsciously planning form, structure and harmonic progression. The momentary and the future abstract are always coexistent and active. Schwartz and Godfrey (1993) define seven principle concepts in contemporary composition: Pitch Logic, Time, Sound Color, Texture, Process, Performance Ritual and Parody (Historicism). Each of these can be considered in terms of both instrument design and mapping strategies for musical interfaces and extend the Norman proposal towards a more detailed musical focus..

Oritz Pérez, Knapp and Alcorn (2007) also discuss this issue when developing a composition-driven approach to using novel interfaces in Díamair for choir and integral music controller. The diagrams in Figure 4 illustrate the way in which the compositional decisions influence the interface and in turn how the interface design influences the instrument design (software synthesis) and compositional decisions.

A combination of these considerations is brought to bear when developing a musical interface that also addresses composition constraints. One example is the author's work with Michael Atherton in their ensemble SynC. The compo-

sition titled Encounter[1], for Hurdy-Gurdy and live electronic processing from the Parallel Lines CD (Paine & Atherton, 2006), utilised the Capybara/Kyma system[2] for live electronic processing of the Hurdigurdi sound, and implemented the WiiMote as the control interface.

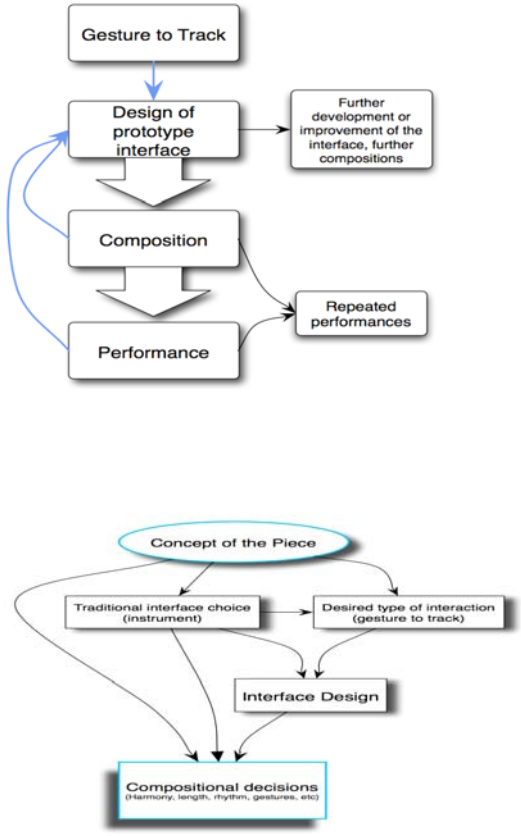


Figure 4 *Composition driven approach to using novel interfaces (above and below)*

Encounter – an example of the WiiMote as musical controller

In developing the synthesis mapping for the WiiMote in this musical work, a number of compositional determinates were considered. The composition contains a large number of synthesis variables. It was not possible to constrain the variable in such a manner that they could all be controlled in realtime. The author chose to automate some variables and leave the following under realtime control; filter center frequency, the triggering of buffer recordings (four buffers), the rate/density of granulation, the triggering of a sample playback (one pre-recorded sample is used in the work), the control of two frequency/pitch variables, the sample-and-hold rate of these oscillators and the delay feedback time also associated with this oscillator instrument which is the solo electronic instrument in the central section of the work.

WiiMote mapping for the musical work *Encounter* is outlines in Table 2.

Synthesis Variable	**Control Source**
WispCrossSynth	→ Nun.Joy.Y
GrainDesnsity	→ Nun.Joy.X
PlayOn	→ Nun.Accel
RecordBuffer1	→ Nun.Trigger2
RecordBuffer2	→ Nun.Trigger.C
LiveRandomBuffer	→ WiiTrigger
Record3	→ WiiButton A
SampleTrigger	→ WiiAccel
Freeze	→ Nun.Joy.Y (acts on Buffer 1, 2, 3 in the Track&Hold Time Freezer)
ChopA	→ WiiPitch (acts on Buffer 3)
RCGain	→ WiiRoll
RCHoldTime01	→ Nun.Yaw
RCFreqMod	→ Nun.Pitch

Table 2 data mapping for Encounter

Conclusion

The author has illustrated an application of the NintendoWiiMote and NunChuck interface for realtime electroacoustic music performance. The composition Encounter was used to discuss the gestural qualities of the interface, compositional constraints and engagement with an audience in addition to the current mapping strategy. The work is still in progress - no formal conclusion has been reached, however, it is clear that the proliferation of wireless gaming interfaces, notably the Nintendo WiiMote, make these questions more relevant and approachable. This makes the development of a shared theoretical basis for interface design and comparison increasingly important. Gurevich & Treviño's (2007) discussion of a framework for an ecology of musical action concludes that Norman's three levels of processing offer a new currency for describing the experience of music creation [placing] the electronic music interface in context. This framework has three distinct advantages: 1) it admits a broader range of aesthetic concerns; 2) it provides a more meaningful way to 'evaluate' an interface; and 3) it expands the scope for the consideration of novel interfaces (pg.109). It is hoped that the musical instrument control model outlined in Figures 1 & 2 will also contribute to this discourse.

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